Sampling Efficiency and Geometric Sensitivity in Cardiff: A Controlled Porto Inference Study

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Released Porto checkpoints | 64 conditions × 3 seeds | Inference only

Finding. Holding generated segment-level latents fixed, increasing the GPS sampling interval from 20 to 50 gave a median paired-batch speedup of **2.503**× (conditional 95% bootstrap interval: **2.432**–**2.692**×). The pilot passed all eight frozen point-estimate criteria. Core length and step-distance summaries were close, while the turning-angle tail changed. These are execution and geometric diagnostics, not evidence of road validity or improved physical realism.

Framework and independent question

Cardiff [1] first compresses discrete road-segment sequences with an autoencoder, generates segment latents with diffusion, and then generates continuous GPS coordinates with a latent-conditioned diffusion model. Our question was narrower than its benchmark: *how do GPS-stage sampling cost and descriptive geometry change when its first-stage generated conditions are kept fixed?* No model training or new sampling algorithm was introduced.

Verified assets and controlled protocol

The official repository [2] was pinned to revision `1138b1ef0646c18e878608a848dafca018b8a888`. Original Porto diffusion state dictionaries loaded with `weights_only=True` and `strict=True`; no keys were missing or unexpected. Formats were inspected before loading. Unchanged SHA-256 values:

`denoiser0:` `09d66ff627e173de6fb75c9cfe0ae130c12a814c8e75d2ce02696ff1f2fba070`

`denoiser1:` `c6bda8e0a67b969d055f37420e527b70e390fecf35f489994568342cac45c02d`

The first 64 valid attribute rows, in original order, were used with seeds 0, 1 and 2. Stage 1 latents were generated once per seed at interval 20, cached and reused in both GPS arms with identical attributes. The eight-condition smoke test returned latents $(8, 32, 128)$ and GPS $(8, 2, 64)$.

Eight-row batches alternated timing order and restored the same pre-call RNG state per pair; batches advanced from the interval-20 post-call state. CUDA-synchronized timings excluded setup, writing and Stage 1. The repeatability control matched exactly (maximum error 0). Effective $\eta = 1$, original conditioning and FP32 were preserved. Hardware: 24 GB RTX 3090; Python 3.11.2, PyTorch 2.5.1+cu124 (versions differ from the author's recipe).

Measured GPS results: 192 trajectories per interval

Diagnostic	Interval 20	Interval 50
Total GPS wall time (s)	41.797	16.702
Time per trajectory (ms)	217.693	86.992
Finite-coordinate fraction	1.000	1.000
Points outside preprocessing box	2/12,288	0/12,288
Median cumulative geodesic length (m)	3394.026	3377.722
Median consecutive-point distance (m)	49.072	49.153
Pooled p95 consecutive distance (m)	163.048	162.193
Pooled p95 unsigned turning angle (°)	62.224	54.180

Lengths use WGS84 geodesics and the original longitude/latitude denormalization: mean $[-8.619011, 41.15872626]$, standard deviation $[0.02382058, 0.01060059]$. Turns use a local equirectangular projection. Region containment is not a road-validity metric.

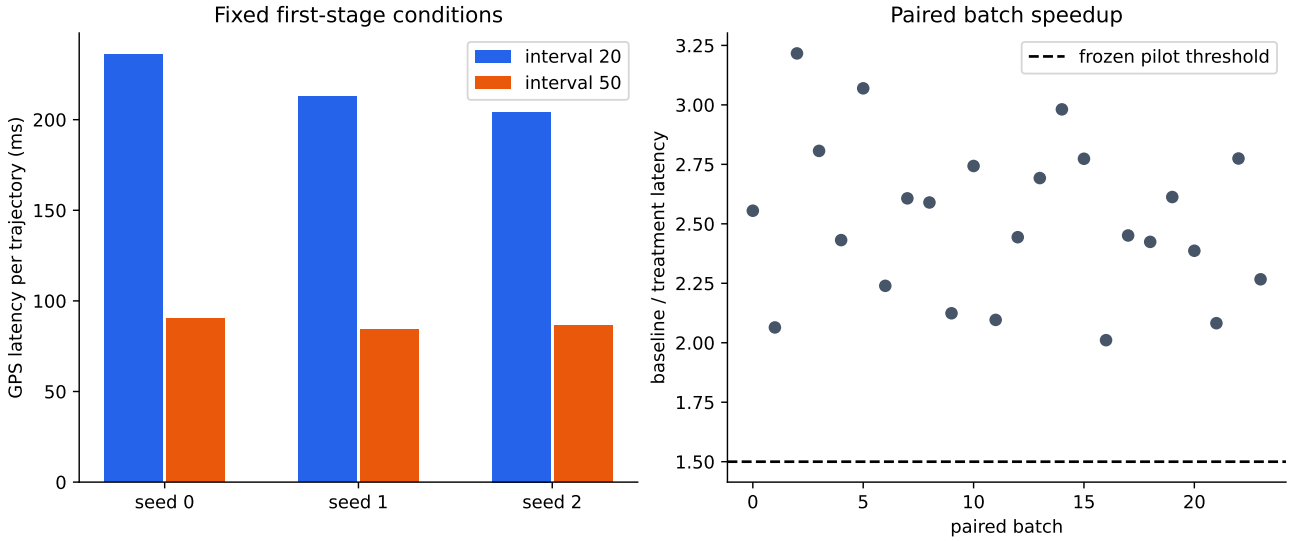


Figure 1. Preserved pilot figure: GPS latency by seed and 24 paired-batch speedups. Stage 1 was fixed within each pair. The dashed line is the frozen 1.5× speedup criterion. No new model outputs or figure data were generated for this note.

Uncertainty and exploratory sensitivity

The median-length ratio (50/20) was 0.99520, conditional 95% interval [0.94165,1.05557]; the pooled p95 step-distance ratio was 0.99475 [0.94610,1.02560]. Both point estimates passed the frozen ±10% margins. All three seeds passed the core geometric margins, although effect directions varied.

The turning p95 ratio was **0.87072** [0.80529,0.92407], a roughly 12.9% reduction. This is **exploratory**: no turning threshold was frozen, and a smaller angle is not established as better. A retained example also changes path shape. Aggregate agreement does not imply identical paths.

Geometry intervals use 2,000 paired attribute-ID resamples retaining all three seeds; timing intervals resample eight batch positions retaining the three seeds. These descriptive intervals are conditional on the ordered subset and RNG coupling. They are not population equivalence tests; only three seeds were studied.

What the RNG pairing controls

The original wrapper sets $\eta = 1$. Interval 20 uses 50 transitions from timestep 980; interval 50 uses 20 from 950. Restoring RNG states shares the unused latent Gaussian, initial GPS Gaussian and transition-noise prefix *by draw index*. Those vectors enter different timesteps and coefficients; common timesteps generally use different draw indices. The comparison therefore combines stochastic schedule effects under one coupling and does not isolate deterministic discretization error.

Clean inference conditioning is intentional

Paper section 3.5.2 prescribes generated latents without noise augmentation at inference, despite augmented training conditions. The helper computes but discards a noisy latent and uses auxiliary noise time 200; the meaning of that convention remains unconfirmed. Original behavior was retained. This is not a conditioning-noise ablation.

Limits and a proposed next question

Cardiff already studies interval efficiency on Singapore [1, section 4.2.3]. This independent Porto pilot introduces no new interval method and does not reproduce benchmark scores. No matching road graph, decoded routes, checkpoint-matched held-out real reference, physical time model or privacy test was evaluated. Attribute split provenance is unknown; box containment and GPS geometry do not establish road validity.

Does faster GPS sampling preserve alignment with the same fixed segment-level road condition? That question requires compatible decoder weights, the original segment-to-geometry mapping, independent held-out controls, asset permissions and a separately approved protocol. Road-alignment work is **proposed and has not been executed**.

References and provenance

[1] Baoshen Guo, Zhiqing Hong, Junyi Li, Shenhao Wang and Jinhua Zhao. *Leveraging the Spatial Hierarchy: Coarse-to-fine Trajectory Generation via Cascaded Hybrid Diffusion.* arXiv:2507.13366v1 (2025). arxiv.org/abs/2507.13366v1.

[2] *Cardiff*, official code and Porto release. github.com/urban-mobility-generation/Cardiff, revision above.

Evidence and summary replay: `https://github.com/ZhijingWu/cardiff-sampling-geometry-audit`. Raw arrays and model weights are excluded. Independent study; no affiliation with or endorsement by the Cardiff authors.